Product Requirements Document

Kano Mobil Distribution: Operations Dashboard and Reporting

Version 0.1 (draft for review) | Owner: Al-Amin B. Yahaya | Status: Pre-build

1. Summary

This product is a web dashboard and reporting tool that gives the Kano Mobil lubricant distribution business one place to manage accounts, orders, deliveries, stock, credit, and payments. It replaces scattered WhatsApp messages, notebooks, and memory with a single record. It exports clean CSV and PDF reports for management, suppliers, and bank or partner conversations, and gives the CEO an overview of the whole operation in under a minute.

The first release targets one depot in Kano and one operator team. It is built to support the expansion to other northern states later without a rebuild.

2. Problem

Credit, stock, and customer relationships are currently tracked in people's heads and in chat threads. This causes four specific losses: receivables grow without anyone noticing, stock runs out or sits too long, the business depends on one person (the founder's father) to know who owes what, and there is no reliable data for the decision to expand. Leadership handover will be impossible without a system of record.

3. Goals and Non-Goals

Goals (release 1)	Non-goals (release 1)
One live view of sales, cash, credit exposure, and stock	Customer-facing ordering app or e-commerce checkout
Every account, order, and payment recorded within 24 hours	Accounting or tax filing (export to the accountant instead)
Overdue and quiet accounts flagged automatically	Multi-depot or multi-state inventory transfers
CSV and PDF export of any table or report	Direct integration with Mobil or supplier systems
Clear margin per litre by product and by segment	Payroll and staff management

4. Users and Roles

Role	Who	Needs	Access
CEO / Owner	Al-Amin	Overview, trends, exports, approvals	Full access, including credit limits and reports
Sales lead	Field sales staff	Add accounts and orders, see own accounts, check balances	Create and edit accounts and orders; no credit limit changes
Depot / dispatch	Store keeper	Stock in and out, delivery list	Stock and delivery screens only
Finance / cashier	Payments clerk	Record payments, see receivables	Payments and receivables; read-only on accounts
Viewer	Father or advisors	Read-only summary	Summary and exports, no edits

5. Core Features (Release 1)

5.1 Overview dashboard. The landing screen shows, for the selected period: total sales (NGN), gross margin (NGN and percent), cash collected, outstanding receivables with aging buckets (0-30, 31-60, 61+ days), total credit exposure against approved limits, litres sold by product, active accounts this week and this month, and accounts flagged as quiet (no order in 30+ days). Each card links to its detail screen.

5.2 Accounts. A customer record holds name, area of Kano, contact, segment (Fleet, Workshop, Reseller), introduced-by source, payment type (Cash or Credit), credit limit, status (Active, On hold, Closed), and notes. Each account page shows its order history, payments, balance, and days since last order. Adding a credit account requires a credit limit before the first order can be placed.

5.3 Orders and invoices. An order has an account, date, line items (product, pack size, quantity, unit price), discount, delivery fee, and status (Draft, Confirmed, Delivered, Cancelled). Confirming an order checks the credit limit and warns or blocks if it would be exceeded. Invoices are generated as PDF. Prices come from a product price list maintained in the system and can be overridden with a reason.

5.4 Payments. Payments are recorded against an account and allocated to invoices. Method (cash, bank transfer, POS) and reference are required. Partial payments are supported. The system calculates days overdue per invoice.

5.5 Stock. Stock is tracked by product and pack size at the depot. Entries are recorded for goods received (with supplier batch number), goods dispatched (linked to an order), and adjustments (with reason). The system shows current stock, reorder level, and days of cover per product. Batch records support the anti-counterfeit traceability goal.

5.6 Deliveries. A daily dispatch list groups confirmed orders by area of Kano. Each delivery can be marked delivered with the name of the receiver. Failed deliveries carry a reason.

5.7 Alerts. The system flags: accounts over credit limit, invoices more than 30 days overdue, accounts quiet for 30+ days, products below reorder level, and any price below the list price. Alerts appear on the dashboard and can be sent as a WhatsApp or email summary every Friday.

5.8 Reports and export. Every table (accounts, orders, payments, stock movements, receivables) can be exported to CSV with the current filters applied. Standard reports are available as PDF, with a shareable link or file for WhatsApp: weekly sales and margin summary, receivables aging, account activity, stock position, and monthly P&L for the business (revenue, cost of goods, gross margin, operating expenses entered manually).

6. Key Metrics (What the Dashboard Must Answer)

Metric	Definition	Why it matters
Gross margin per litre	(Selling price minus landed cost) per litre, by product	Shows whether the business is actually making money on each product
Receivable days (DSO)	Outstanding receivables divided by average daily sales, over 30 days	Early warning for cash trouble
Active accounts	Accounts with at least one order in the last 30 days	Shows whether the network is alive
Repeat rate	Share of accounts that ordered again within 60 days of first order	Shows loyalty and the real value of the network
Stock cover	Current stock divided by average daily dispatch, per product	Prevents stockouts and dead stock
Credit exposure ratio	Total outstanding balance divided by total approved limits	Shows how much risk the business is carrying

7. Data Model (Summary)

The core entities are Account, Contact, Product, PriceList, Order, OrderLine, Invoice, Payment, PaymentAllocation, StockMovement, StockBatch, Delivery, and User. All monetary values are stored in NGN with two decimal places. Every record carries created-by, created-at, and updated-at fields. Deletions are soft (status set to Cancelled or Archived) so reports stay accurate.

8. Non-Functional Requirements

Area	Requirement
Mobile first	Works well on Android phones of mid-range spec and on a 3G or patchy 4G connection. Pages load under 3 seconds on a typical Kano mobile connection.
Offline entry	Sales staff can enter an order without signal; it syncs when the connection returns. Conflicts show for review, not silent overwrite.
Security	Login per user, role-based access, audit log for credit limit, price, and payment changes. Passwords hashed; backups daily.
Data ownership	Full CSV export of all business data at any time, so the business is never locked in.
Currency and format	NGN with thousands separators; dates in DD-MMM-YYYY; works in English.
Reliability	Target 99 percent uptime during business hours. Daily automated backup with a tested restore.

9. Suggested Approach and Build Options

Option A, fastest start: a Google Sheets or Airtable base with a mobile form and dashboard. Cost is near zero and it can go live in days, but it becomes fragile above a few hundred accounts and offers weak access control. Option B, recommended: a progressive web app (installable on the phone home screen) with a Supabase database and authentication, a React front end, and built-in CSV and PDF export. This gives proper roles, offline entry, and scale to other states. Option B takes roughly 4 to 6 weeks for release 1 with a small technical team. Recommend starting with Option A for the first 4 weeks of live data if you need to begin immediately, then migrating to Option B with the CSV exports as the bridge.

10. Release Plan

Phase	Scope	Exit criteria
Phase 0 (week 1)	Confirm data fields, price list, credit policy, and roles with the team. Load the existing accounts into the tracker.	Signed-off field list; price list and credit policy written
Phase 1 (weeks 2-4)	Accounts, orders, invoices, payments, receivables aging, CSV export, overview dashboard	Ten live users recording daily for two weeks with no lost orders
Phase 2 (weeks 5-7)	Stock, deliveries, alerts, PDF reports, WhatsApp summary	Stock counts reconcile with physical count within 2 percent
Phase 3 (weeks 8-10)	Offline entry, margin analysis, P&L report, role refinement, backup and restore test	Monthly close produced from the system without manual rework
Later	Multi-depot, additional northern states, supplier portal	Decided after Kano meets its exit criteria

11. Success Criteria

Within 90 days of go-live: every order and payment is recorded in the system (verified by weekly reconciliation against cash and bank records), receivables above 60 days fall by at least half from the baseline, stock-outs on the top five products drop to zero, and the CEO can produce the weekly summary in under 15 minutes without asking staff.

12. Risks and Mitigations

Risk	Mitigation
Staff don't adopt the system and keep using chat	Make the system the only place payments and credit changes are approved; train on the phone; keep entry under two minutes per order
Data entered inaccurately	Required fields, dropdowns, daily reconciliation of cash and orders, audit log
Credit limits ignored under pressure from family relationships	Limits enforced by the system; overrides require CEO approval and a reason
Connectivity failures	Offline entry in phase 3; Option A bridge until then
Key-person risk (data lives in one person's head)	Migration of all existing account knowledge into the system in phase 0

13. Open Questions

Which Mobil products and pack sizes are in scope for the first price list? What is the agreed credit policy and who approves limits? Does the business need VAT and WHT handling on invoices? Which bank accounts and POS providers must payments reconcile against? Who on the team will own day-to-day admin of the system after launch? Should WhatsApp be used for order intake, and if so, will staff copy those orders into the system or will we build an intake step?

14. Decisions Needed Before Build

Approve the release 1 scope in section 5. Choose Option A or Option B in section 9. Confirm the roles in section 4 and who holds each. Sign off the credit policy and price list, since the system enforces both.